

Aiden Lowery

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EDUCATION

The University of Texas at Austin

Expected May 2029

Bachelor of Science in Computer Science and Mathematics

Austin, TX

Relevant Coursework: Data Structures, Discrete Mathematics, Multivariable Calculus.

Entered with sophomore standing from AP credits — on track to graduate in three years.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, MicroPython, HTML/CSS
- **Machine Learning & Data:** PyTorch, OpenCV, YOLO, CUDA, Pandas, NumPy
- **Web & Frameworks:** Django, REST APIs
- **Tools & Platforms:** Git, GitHub, Linux, Claude Code, GitHub Copilot
- **Hardware:** Raspberry Pi, Raspberry Pi Pico, sensor integration, circuit soldering

PROJECTS

CubeArena — Competitive Speedcubing Platform | *Python, PyTorch, YOLO, OpenCV, CUDA* 2025 – Present

- Architected a two-model computer-vision pipeline (YOLO cube detector + ResNet-18 move classifier on temporal frame differences) to recognize cube moves from live video.
- Designed a decoder to backtrack from known states in order to clean up inevitable inaccuracies in computer vision results.
- Built the training pipeline from OpenCV prototyping to a fine-tuned YOLO model, using a Bluetooth smart cube (bleak) to capture ground-truth labels.

Algorithmic Trading & Valuation Models | *Python* 2025

- Built Python security trading algorithms that produced positive returns in backtests and small live trials.
- Developed equity valuation models, such as statistical arbitrage, and a lightweight framework to run live financial simulations without large third-party backtesting platforms.

Autonomous Quadcopter Drone | *Raspberry Pi, MicroPython, Mission Planner* 2024 – 2025

- Designed, built, and programmed a modular quadcopter from scratch — custom 3D-printed frame, hand-soldered electronics, and autonomous flight via Mission Planner + MicroPython on a Raspberry Pi.

Small-Business Website (Freelance Contract) | *Python, Django, HTML/CSS* 2022

- Delivered and deployed a full-stack website for a construction business as a paid contract, building both front and back end with custom forms and a photo gallery.

LEADERSHIP & WORK

Walgreens, Team Member — Friendswood, TX 2024 – 2026

- Greeted and maintained Relationships with customers; Responsibly handled transactions and photo printing while problems solving errors that come up.

Varsity Cross Country & Track, Team Captain — Friendswood High School 2022 – 2026

- Led and mentored 40+ teammates; NHSTCA Cross Country All-American (2025) and Texas All-State.

HONORS

- National Merit Commended Scholar; AP Scholar with Distinction; UT CS Academy for All: ML (merit, 2024).